datacare. limited

SECURITY · ENGINEERING · ASSURANCE

WEB
www.datacare.co.ke
EMAIL
info@datacare.co.ke
PHONE
0722 155 752

DOCUMENT 1 OF 2 — PROGRESS REPORT

Elimu Pay — Production-Hardening Progress Report

Prepared by: Datacare Limited **For:** David Njiru, Elimu Pay **Date:** 17 June 2026 **Contact:** info@datacare.co.ke · www.datacare.co.ke · 0722 155 752

1. Executive summary

Work to take Elimu Pay from a working build to a launch-ready, production-grade platform is well advanced and **on track for the agreed launch window**. We now have full access to the codebase and database, and have completed the security, stability, and core "market-readiness" work — the items a school and a regulator would expect of a system that handles real money and the personal data of children.

Every change has been delivered into a single, reviewed change-set, and each piece has been tested against a large, realistic copy of the data (thousands of students and payments) before being accepted. The platform is materially more secure, faster under load, and now supports the payment and communication features that schools asked for.

A small amount of work remains — chiefly a visual/usability refresh and a few items that depend on information only you or the schools can provide (payment and messaging account details). These are listed in Sections 5 and 6.

2. What we have completed

Security and data protection

- Closed the highest-risk security gaps around **payment integrity and login**, so that

fake payments cannot be injected and login/2-factor sessions cannot be forged.

- Strengthened how sensitive data is stored and how automated/background jobs are

authorised, and removed a diagnostic endpoint that should never reach production.

- Independent penetration-test findings supplied earlier were reconciled against the

live system; the outstanding high-risk items have been addressed.

Reliability and scale

- Re-engineered the database connection approach so the platform stays responsive as the

number of schools grows (essential for a multi-school, cloud deployment).

- Added an automated safety net of **38 tests** plus a large-scale stress test that we run

before accepting any change, so regressions are caught early.

Fee collection and compliance

- **Tax-compliant invoice numbering.** Invoices now carry unique, sequential numbers

assigned by the system — a requirement for KRA-compliant record-keeping. This was proven correct even when many invoices are produced at the same moment.

- **Instant M-Pesa payment requests (STK Push).** Staff can now prompt a parent's phone

to pay directly; a confirmed payment is recorded and reconciled to the right student automatically, and the parent receives a confirmation.

- **Faster, server-generated invoices.** Invoice PDFs are now produced by the server

rather than the staff member's browser, so sending invoices — including in bulk — no longer slows down or freezes the screen.

Parent communication

- **Real SMS fee reminders** via a Kenyan bulk-SMS gateway, alongside the existing email

and WhatsApp options, including an automated scheduled reminder run.

Performance and pre-launch tidy-up

- Fixed the heaviest screen in the app (the reminders list) so it stays fast even with

thousands of students.

- Resolved a class of display/console warnings that affected polish and correctness.
- Removed the public "demo" area ahead of the live launch.

---

3. Testing performed

Testing was done against a **large, realistic dataset** — roughly **5,000 students across ~600 schools, with 3,000 payments and 2,000 invoices** — to mimic real load rather than a small sample.

Area tested	How	Result
Invoice numbering under pressure	Issued 75 invoices simultaneously	All unique, sequential, none duplicated; re-sending an invoice kept its original number
Instant payment (STK) security	Probed the payment endpoints directly	Staff-only access enforced; the payment-confirmation channel safely rejects forged calls without recording anything
Server-generated invoices	Issued and downloaded invoices end-to-end	Produced correctly and quickly; a failed email no longer voids an issued invoice
SMS and payment features when not yet configured	Triggered them without account details	Fail gracefully with a clear message — never crash
Speed under load	Loaded the main screens with the full dataset	Key list screens load in a fraction of a second when paged; the heaviest screen was reduced dramatically
Automated regression suite	38 automated checks on every change	All passing

---

4. Key findings

- **The new fee-collection and compliance features are correct and safe.** The most

important property — that money and invoice numbers are handled accurately even under simultaneous activity — was demonstrated, not assumed.

- **Security posture is strong.** Payment-confirmation channels reject anything they

cannot verify, and sensitive actions are limited to authorised staff.

- **Performance is good and will be better in production.** The biggest gains came from

only loading what is shown on screen; one heavy screen has been brought into line with the rest. (Measured speeds during testing are slower than the live system will be, because the test environment runs in an un-optimised development mode.)

- **A few usability/polish items remain** and are itemised below; none are blockers.

---

5. What remains

1. **Visual and usability refresh** — a consistent design pass and accessibility improvements across the screens. We have deliberately held back front-end changes pending your written go-ahead; this is the main remaining body of work and can begin on your word.
2. **Two small front-end additions** already approved — an in-app button to trigger an M-Pesa payment request, and a one-click invoice download — to be wired next.
3. **Final compliance items** — a published data-protection/privacy position aligned to Kenya's Data Protection Act.

---

6. What we need from you / the schools

To switch the new features fully "on" in production, we will need:

- **Bulk-SMS account** (sender name) so SMS reminders can send under the school's identity.
- **M-Pesa details per school** — each school's own paybill/till and the matching M-Pesa

online passkey. Per your direction, every school uses its **own** M-Pesa number for its collections; the platform is built around that model.

- **Email sending credentials** for invoice and payment-confirmation emails.

We can provide a short, plain checklist for each of these.

---

7. Outlook

The platform is in a strong, launch-ready position on security, stability, compliance, and core features. The remaining work is primarily the visual/usability refresh and switching on the third-party accounts above. We remain confident in the agreed launch window and will share a final go-live checklist as we approach it.

This report summarises progress and outcomes. Technical implementation detail is held in the project's source control and is available to your team on request.

— **Datacare Limited** info@datacare.co.ke · www.datacare.co.ke · 0722 155 752

datacare. limited

SECURITY · ENGINEERING · ASSURANCE

WEB
www.datacare.co.ke
EMAIL
info@datacare.co.ke
PHONE
0722 155 752

DOCUMENT 2 OF 2 — GO-LIVE SETUP GUIDE

Elimu Pay — Go-Live Setup Guide

Prepared by: Datacare Limited · **For:** David Njiru, Elimu Pay · **Date:** 17 June 2026 **Contact:** info@datacare.co.ke · www.datacare.co.ke · 0722 155 752

This guide walks you through everything that must be set up on your side to move Elimu Pay into the live (production) environment. Each section says **what to do**, the **rough cost**, and **exactly what to send back to us** so we can connect it.

> **Prefer us to handle all of it?** If you'd rather not do these steps yourself, Datacare can take care of the entire production setup end-to-end — registering and configuring every account (domain, hosting, database, email, SMS, M-Pesa), wiring it all up, and deploying — for a **one-time setup fee of KES 20,000**. This is a separate, optional service and is **not part of the original arrangement**. (Third-party charges above — domain, hosting, SMS units, etc. — are billed by those providers and are not included in this fee.)

> **Important — ownership.** Please create each account **in Elimu Pay's name** (your business email and your payment card). You own the accounts and the billing; we only need the access keys to connect them. Where an account supports it, you can instead add us as a team member (we'll give you the email to invite). Never let a vendor account be owned by an individual contractor.

> **How to send us credentials — securely.** Do **not** send keys over plain email, SMS, or WhatsApp. Use one of these:
>
> 1. A one-time secret link — go to **onetimesecret.com**, paste the value, send us the link (it self-destructs after we open it), or
> 2. Share via a password manager (we can use a shared 1Password/Bitwarden vault), or
> 3. Read them to us on a call while we enter them.

At a glance — the pieces

Piece	Provider (recommended)	Purpose	Indicative cost
Domain name	**simbahost.co.ke** (.co.ke)	Your web address	KES 1,500/yr
Hosting	Vercel *(already set up)*	Runs the application	Currently free tier; ~$20/mo Pro for production scale
Database	Neon — PostgreSQL *(already set up)*	Stores all data	Currently free tier; ~$19/mo for production scale
Email	**Resend**	Invoices, reminders, confirmations	Free up to 3,000/mo; ~$20/mo for 50,000
SMS	Celcom Africa	SMS fee reminders	Pay-as-you-go, ~KES 0.30–0.80 per SMS
M-Pesa	Safaricom Daraja	Payments & instant pay prompts	Free API; paybill per Safaricom

Already in place. The application is built and **live on Vercel** (currently at https://feetracker-seven.vercel.app) with its database on **Neon** — so hosting and the database need nothing from you. The only thing to note is that, for production scale, we may move both to their **paid tiers** (Vercel Pro ~$20/mo, Neon ~$19/mo) for performance and reliability as schools come on board; we'll advise when that's worthwhile. Everything else below can start on a **free tier**.

Step 1 — Buy a domain name

What to do

1. Decide the address. We suggest **elimupay.co.ke** (Kenyan, trusted locally).
2. Register it at **simbahost.co.ke** — search for the name and check out (**KES 1,500/year**).
3. After purchase, you'll get a login to the registrar's **DNS / domain management** page. We will give you a short list of DNS records to add (for the website, email, etc.), or you can add us as a user on the registrar so we configure it.

What to send us

- The domain name(s) you bought, and **access to the DNS settings** (invite us, or be

available to paste in the 4–6 records we provide).

We connect your new domain (Step 1) to the existing Vercel deployment so the app answers at `https://elimupay.co.ke` instead of the temporary address.

Step 2 — Set up email sending (Resend)

This powers invoice emails, payment confirmations, and email reminders.

What to do

1. Go to **resend.com** → **Sign Up** (free tier: 3,000 emails/month, 100/day).
2. In the left menu, open **Domains** → **Add Domain** and enter your domain (e.g. `elimupay.co.ke`). Resend will show you **3–4 DNS records** (SPF, DKIM, and a return-path).
3. Add those records in your domain's DNS page from Step 1 (or send them to us to add). Wait for Resend to show the domain as **Verified** (usually minutes, sometimes a few hours).
4. Open **API Keys** → **Create API Key**, name it "Elimu Pay Production", and copy the key (it starts with `re_…`). **You can only see it once** — copy it immediately.

What to send us (securely)

- The **Resend API key** (`re_…`).
- Confirmation that your domain shows **Verified** in Resend.
- The **"from" address** you want parents to see (e.g. `fees@elimupay.co.ke` or

`no-reply@elimupay.co.ke`).

> Note: the app currently emails via a Google account. We will switch it to Resend (a small change on our side) so that, once you send the key above, everything sends under your own domain with better deliverability.

Step 3 — Set up SMS (Celcom Africa)

What to do

1. Visit **celcomafrica.com** and **register for a Bulk SMS account** (or email/call their sales team — they often onboard you directly).
2. Ask them to **register a Sender ID** for you — a short text name parents will see as the sender, e.g. `ELIMUPAY` or your school brand. (Sender-ID approval goes through the Communications Authority and can take **1–2 weeks**, so start this early. There is a small one-time registration fee.)
3. **Buy SMS units** to load your account (start small, e.g. KES 2,000–5,000).
4. From your Celcom account dashboard, obtain your **API key**, **Partner ID**, and the approved **Sender ID (shortcode)**.

What to send us (securely)

- **API key**, **Partner ID**, and the approved **Sender ID**.

Until these arrive, the SMS feature stays safely dormant (it simply does nothing).

Step 4 — Set up M-Pesa (Safaricom Daraja) — per school

Per your direction, **each school collects to its own M-Pesa paybill/till**. There are two parts: a one-time developer setup (platform level), and a per-school detail.

A. One-time (platform) — Daraja developer account

1. Go to **developer.safaricom.co.ke** → **Sign up** with the Elimu Pay email.
2. Create an **App** (it gives you a **Consumer Key** and **Consumer Secret**). These let the platform talk to M-Pesa on schools' behalf.

B. Per school — their M-Pesa details For each school joining the platform you (or the school) will need:

1. The school's **Paybill or Till number** (their Lipa na M-Pesa business shortcode).
2. The school's **M-Pesa "online" passkey** for that shortcode (Safaricom issues this when the shortcode is enabled for online payments / "Lipa na M-Pesa Online"). This is required for the **instant payment prompt (STK Push)** feature.
3. We register the school's paybill inside the app so real-time payment notifications flow in.

Going live: Daraja starts in a **sandbox** (test) mode. When you're ready, Safaricom runs a short **go-live** approval to switch your app to production. We handle the technical side; you may be asked to confirm business details.

What to send us (securely)

- The **Consumer Key** and **Consumer Secret** from your Daraja app.
- For each school: **Paybill/Till** and the **online passkey**.

Behind-the-scenes secrets (we handle these)

A few internal secrets (for encrypting data, securing automated jobs, and authenticating M-Pesa callbacks) are generated by us and stored only in the hosting environment. **You don't need to do anything here** — they're already set in the current deployment.

Master checklist — what to send us

#	Item	From
1	Domain name + DNS access	Step 1
2	Resend API key + verified domain + "from" address	Step 2
3	Celcom API key + Partner ID + Sender ID	Step 3
4	Daraja Consumer Key + Secret; per-school Paybill + passkey	Step 4

Send each item via a one-time secret link (see top of this guide). Hosting and the database are already running, so as soon as we have items 1–2 the live site is on your own domain with email sending; SMS (3) and M-Pesa (4) switch on as those arrive.

Suggested improvements beyond the current scope

You asked for ideas to improve the application further. These are **optional** and **not part of the current engagement** — we've ranked them by value and added a rough effort guide so you can decide what to commission next. We're happy to quote any of them firmly on request.

Recommended first — protect the live system

Improvement	What it gives you	Rough effort
Error & uptime monitoring	Instant alerts the moment something breaks or the site goes down, plus a dashboard of issues — essential peace-of-mind for a system handling real money.	Small (~1–2 days)
Automated backups + tested restore	Daily off-site backups *and* a proven restore, so school and payment data can never be lost — we verify the restore works, not just that a backup exists.	Small (~1–2 days)

High value — grow collections & cut workload

Improvement	What it gives you	Rough effort
Parent-facing balance & pay page	A link each parent can open to see their child's balance and pay on the spot — fewer calls to the bursar and faster collections.	Medium (~1–2 weeks)
Payment reconciliation dashboard	A daily view of collected vs. expected, per class/school, with unmatched payments flagged for action.	Medium (~1 week)
Per-school onboarding wizard	New schools self-onboard their paybill, students and fees with minimal hand-holding — lets you scale sign-ups without manual setup.	Medium (~1–2 weeks)

Nice to have — richer engagement & reporting

Improvement	What it gives you	Rough effort
Official WhatsApp Business API	Fully automated WhatsApp reminders (not click-to-send) with delivery receipts.	Medium + WhatsApp approval
Role-based reporting & exports	Termly collection reports for principals and boards, exportable to Excel/PDF.	Small–Medium

---

What happens next on our side

- We switch the email engine to **Resend** so it's ready the moment your key arrives.
- We wire the two approved front-end additions: the **"Request M-Pesa payment"** button and

one-click invoice download.

- On your go-ahead, we begin the **visual/usability refresh** (the main remaining work).
- We point your new domain at the existing Vercel deployment and run a final go-live checklist

with you (including any move to the paid hosting/database tiers).

We're happy to hop on a short call and do Steps 1–2 together live if that's easier.

— **Datacare Limited** info@datacare.co.ke · www.datacare.co.ke · 0722 155 752